Project Report

Project Title: Smart Cultural Storyteller

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Introduction

The Smart Cultural Storyteller is an AI-powered project designed to digitally revive cultural, mythological, and folk stories. It uses AI-based text generation, image creation, and audio narration to present stories in an engaging and interactive format. The platform aims to make traditional storytelling more accessible and enjoyable for modern audiences.

Problem Statement

Many cultural and folk stories are fading from memory as oral storytelling traditions decline. Younger generations are more inclined toward digital media, which creates a gap between cultural heritage and modern engagement. This project attempts to bridge that gap by using AI to retell these stories in visually and interactively compelling ways.

Objectives

- Develop an AI system capable of generating stories in multiple languages.
- Convert generated stories into meaningful visuals using AI image generation.
- Provide audio narration for an immersive storytelling experience.
- Deliver an interactive platform that feels like a digital comic-book experience.

Methodology

The project followed a structured approach involving requirement gathering, system design, and implementation of AI-based modules. The major steps included:

- Story generation using Google Gemini API.
- AI-based image generation using Stable Diffusion and related models.
- Audio narration powered by text-to-speech APIs.
- Frontend integration and interactivity using Streamlit.
- Enhancing usability with loaders and session state management.

Implementation

The project was developed using Python with modular architecture:
- app.py: Main Streamlit interface.
- utils.py: Helper functions for story, image, and audio generation.
- .env: Secure storage of API keys.

Challenges included handling API errors, managing multiple languages, ensuring images generate only when requested, and maintaining session persistence.

Results & Observations

The Smart Cultural Storyteller successfully demonstrated:
• Story generation in multiple languages.
• AI-based image creation aligned with story context.
• Immersive storytelling with narration.
• Comic-book-like interactive user experience.

Observation: While the system worked well, image quality consistency was a challenge. Loader integration and session management improved the usability significantly.

Conclusion & Future Work

This project demonstrated the potential of AI to preserve and promote cultural heritage through modern, interactive storytelling. Future improvements could include:
• Adding background music and sound effects.
• Improving image consistency and realism.
• Deploying on cloud platforms for wider reach.
• Allowing users to contribute their own cultural stories.